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Subject Computer Networks

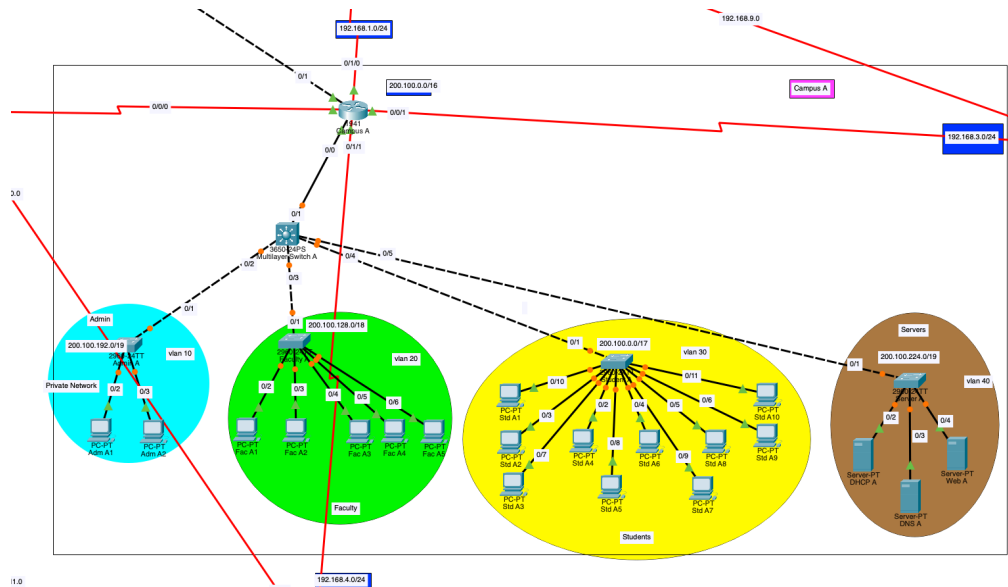
Final Project



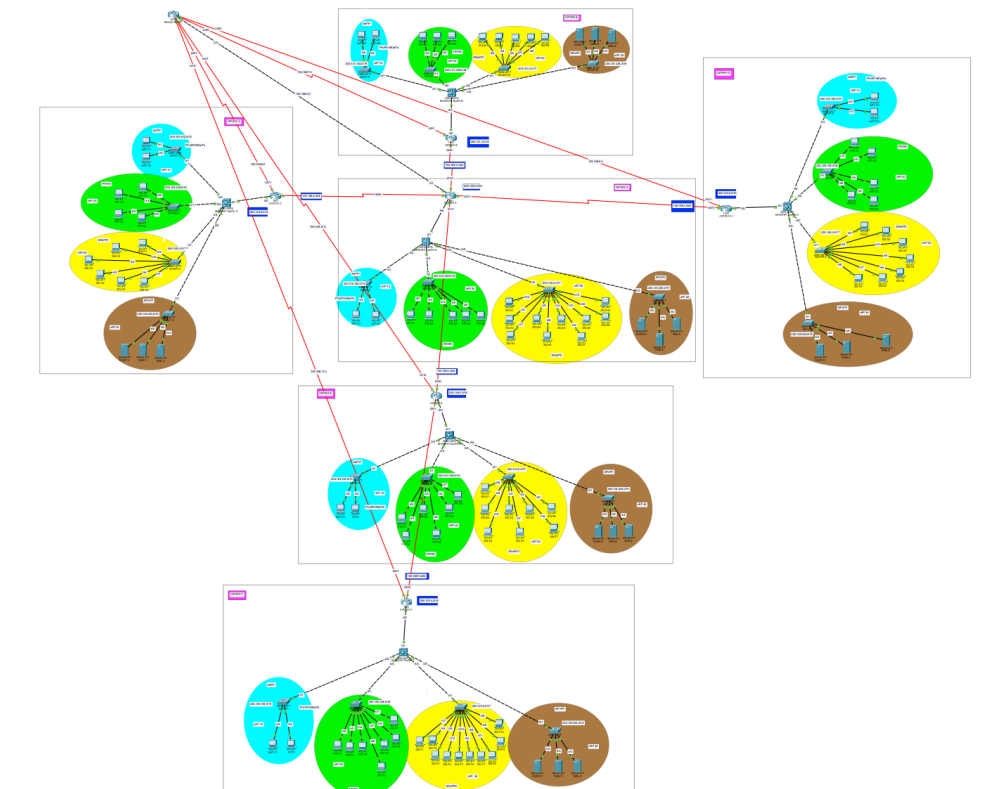
Report

This is Campus A topology.

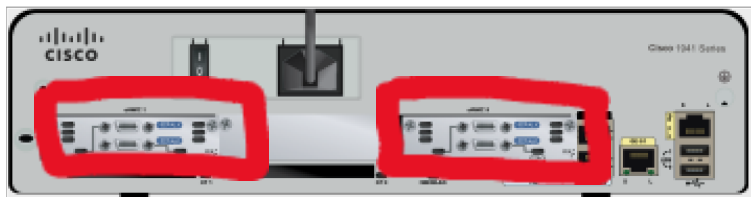
Note: All other 5 campuses topologies are similar.



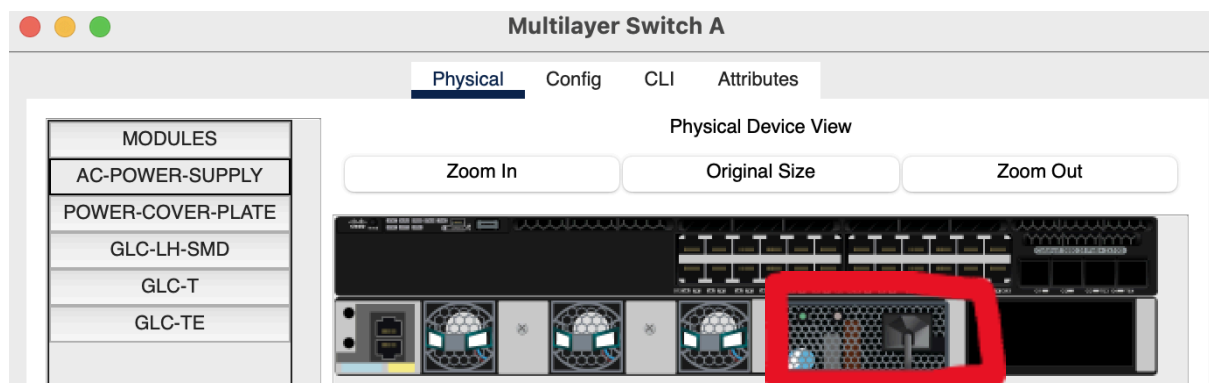
Topology of all 6 campuses.



In router 1941, I had added serial ports to connect all routers to each other.



3650 Multilayer switch to which all department switches of 1 campus is attached. I had also apply power supply in order of its working.



IP Addressing & Subnetting Table - Including VLSM plans

For Campus A: 200.100.0.0/16, for Campus B: 200.101.0.0/16 and so on for all campuses.

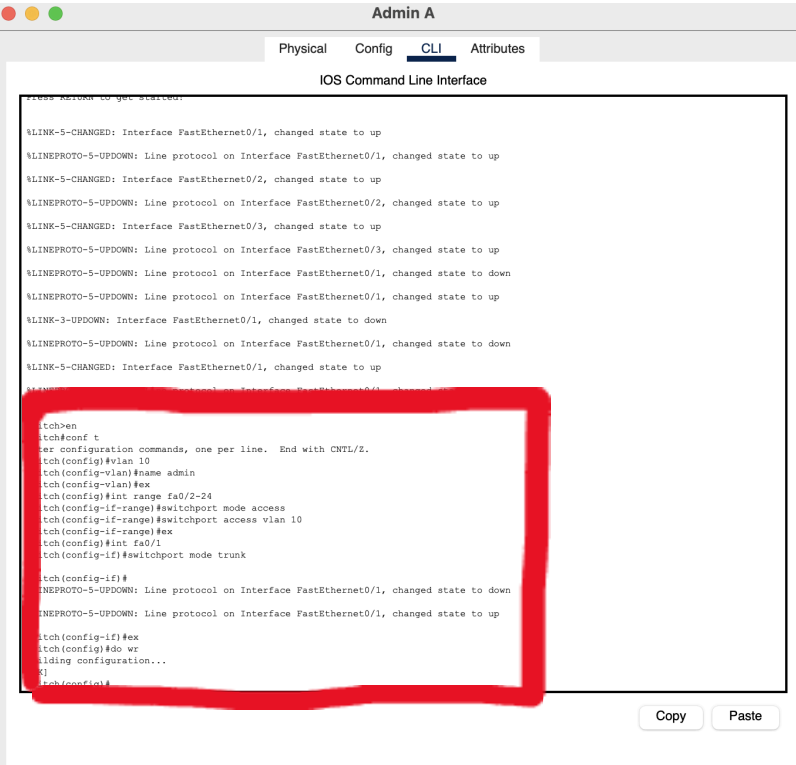
For each department:

Department	Campus A	Campus B	Campus C	Campus D	Campus E	Campus F
Admin	200.100.192.0	200.101.192.0	200.102.192.0	200.103.192.0	200.104.192.0	200.105.192.0
Faculty	200.100.128.0	200.101.128.0	200.102.128.0	200.103.128.0	200.104.128.0	200.105.128.0
Students	200.100.0.0	200.101.0.0	200.102.0.0	200.103.0.0	200.104.0.0	200.105.0.0
Servers	200.100.224.0	200.101.224.0	200.102.224.0	200.103.224.0	200.104.224.0	200.105.224.0

For Campus A to Campus B: 192.168.1.0/24, for Campus A to Campus C: 192.168.2.0 and so in the same for all other campuses for inter communication.

Also all connection of campuses to Backup server uses the same technique of IP allocation like for Backup Server and Campus A: 192.168.6.0/24 and so on for all other campuses.

Vlan for Admin: (Campus A)



Admin A

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

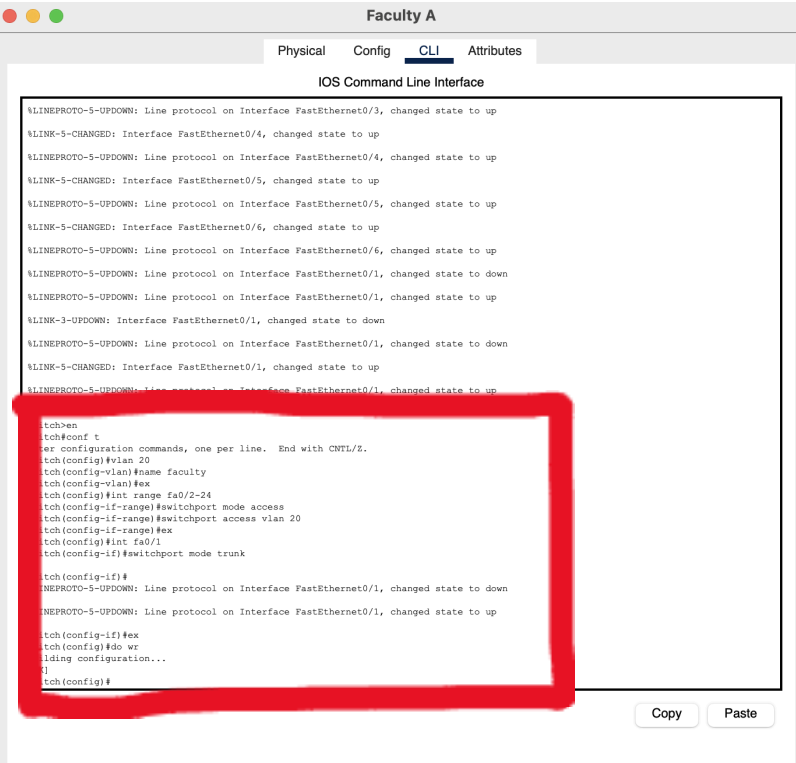
tch>en
tch#conf t
er configuration commands, one per line. End with CNTL/Z.
tch(config)#vlan 10
tch(config-vlan)#name admin
tch(config-vlan)#ex
tch(config)#int range fa0/2-24
tch(config-if-range)#switchport mode access
tch(config-if-range)#switchport access vlan 10
tch(config-if-range)#ex
tch(config)#int fa0/1
tch(config-if)#switchport mode trunk

tch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

tch(config-if)#ex
tch(config)#do wr
iding configuration...
[]
tch(config)#
```

Copy Paste

Vlan for Faculty: (Campus A)



Faculty A

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

tch>en
tch#conf t
er configuration commands, one per line. End with CNTL/Z.
tch(config)#vlan 20
tch(config-vlan)#name faculty
tch(config-vlan)#ex
tch(config)#int range fa0/2-24
tch(config-if-range)#switchport mode access
tch(config-if-range)#switchport access vlan 20
tch(config-if-range)#ex
tch(config)#int fa0/1
tch(config-if)#switchport mode trunk

tch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

tch(config-if)#ex
tch(config)#do wr
iding configuration...
[]
tch(config)#
```

Copy Paste

Vlan for Students: (Campus A)

Student A

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name students
Switch(config-vlan)#ex
Switch(config)#int range fa0/2-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

CopyPaste

Vlan for Servers: (Campus A)

Server A

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

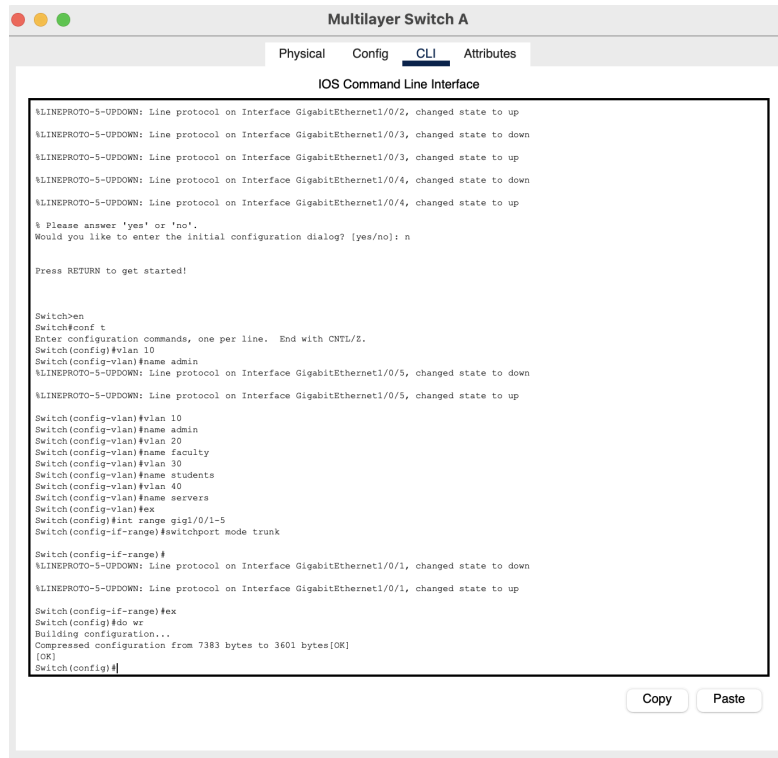
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name servers
Switch(config-vlan)#ex
Switch(config)#int range fa0/2-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#ex
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#ex
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

CopyPaste

Vlan Configuration at Multilayer Switch:



Gateways configuration for each vlan at router:



Note: This (vlan + gateway) is shown for only Campus A. This same way goes for all other campuses.

Set bandwidth of primary higher and backup link slower to make sure eigrp always first ensure primary link.

Campus A

Physical Config CLI Attributes

IOS Command Line Interface

```
2 192.168.4.1/24 is directly connected, Serial0/1/1
D 192.168.5.0/24 [90/2681856] via 192.168.4.2, 00:13:55, Serial0/1/1
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks

Router(config)#int se0/1/0
Router(config-if)#bandwidth 100000
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.1.2 (Serial0/1/0) is down: interface down

Router(config-if)#int se0/1/0
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.1.2 (Serial0/1/0) is up: new
Router(config-if)#int se0/0/1
Router(config-if)#bandwidth 100000
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.3.2 (Serial0/0/1) is down: interface down

Router(config-if)#bandwidth 100000
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.3.2 (Serial0/0/1) is up
Router(config-if)#int se0/1/1
Router(config-if)#bandwidth 100000
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.4.2 (Serial0/1/1) is down: interface down

Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.4.2 (Serial0/1/1) is up: new adjacency

Router(config-if)#int se0/0/0
Router(config-if)#
Router(config-if)#bandwidth 100000
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.2.2 (Serial0/0/0) is down: interface down

Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.2.2 (Serial0/0/0) is up: new adjacency

Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#int gig0/1
Router(config-if)#bandwidth 10000
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.6.1 (GigabitEthernet0/1) is down: interface down

Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.6.1 (GigabitEthernet0/1) is up: new adjacency
|
```

Copy Paste

EIGRP at Campus A router:

Campus A

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

Router#enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
%SYS-5-CONFIG_I: Configured from console by console
no network 192.168.1.0
Router(config-router)#no network 200.100.0.0
Router(config-router)#no network 200.101.0.0
Router(config-router)#ex
Router(config)#router eigrp 10
Router(config-router)#network 192.168.1.0 255.255.255.0
Router(config-router)#network 192.168.2.0 255.255.255.0
Router(config-router)#network 192.168.3.0 255.255.255.0
Router(config-router)#network 192.168.4.0 255.255.255.0
Router(config-router)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 192.168.1.2 (Serial0/1/0) is up: new adjacency

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 10
Router(config-router)#network 200.100.0.0 255.255.0.0
Router(config-router)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
|
```

Copy Paste

Note: This eigrp is shown only for Campus A. Similary for all other campuses is also implemented.

DHCP pools for Campus A:

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

Physical

Config

Services

Desktop

Programming

Attributes

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

Start IP Address

200

100

224

0

Subnet Mask

255

255

224

0

Maximum Number of Users

0

TFTP Server

0.0.0.0

WLC Address

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serversPool	200.100...	200.100...	200.100...	255.255...	5...	0.0.0.0	0.0.0.0
studentsPool	200.100...	200.100...	200.100...	255.255...	3...	0.0.0.0	0.0.0.0
facultyPool	200.100...	200.100...	200.100...	255.255...	1...	0.0.0.0	0.0.0.0
adminPool	200.100...	200.100...	200.100...	255.255...	5...	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	200.100...	255.255...	0	0.0.0.0	0.0.0.0

DHCP broadcast message request is set for specifically sending DHCP request to that particular server.

Physical

Config

CLI

Attributes

IOS Command Line Interface

Setting Configuration...

[OK]

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router con0 is now available

Press RETURN to get started.

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int gig0/0.10

Router(config-subif)#ip helper-address 200.100.224.3

Router(config-subif)#int gig0/0.20

Router(config-subif)#ip helper-address 200.100.224.3

Router(config-subif)#int gig0/0.30

Router(config-subif)#ip helper-address 200.100.224.3

Router(config-subif)#int gig0/0.40

Router(config-subif)#ip helper-address 200.100.224.3

Router(config-subif)#ex

Router(config)#do wr

Building configuration...

[OK]

Router(config)#

Copy

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In DNS service, address of Web server is given.

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

Physical

Config

Services

Desktop

Programming

Attributes

DNS

DNS Service

On

Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	campusa.com	A Record	200.100.224.4

DNS Cache

Index.html of Web server is updated for web services of Campus A.

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

Physical

Config

Services

Desktop

Programming

Attributes

Web A

File Name: index.html

<html>

<center>Campus A</center>

<hr>Welcome to Campus A.

<p>Quick Links:

A small page

Copyrights

Image page

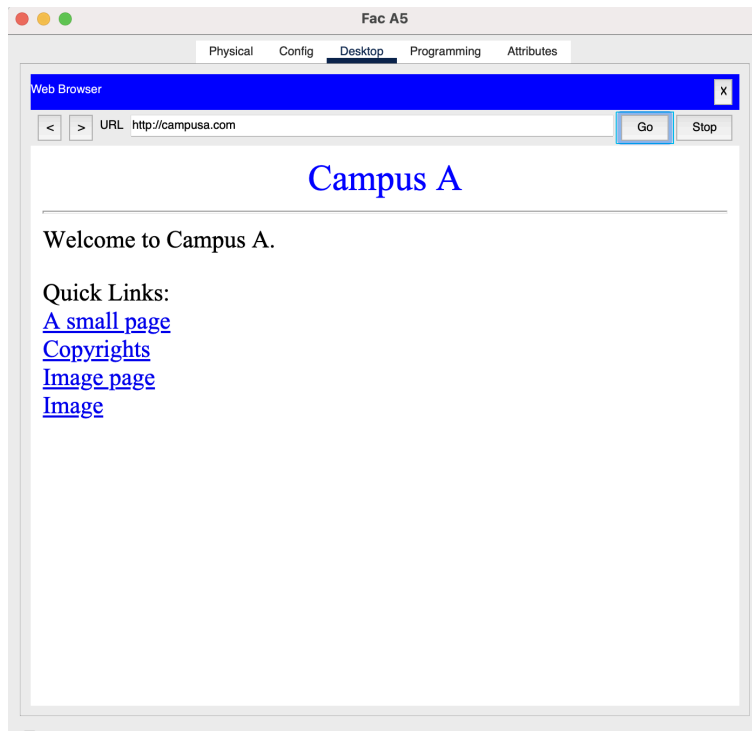
Image

</html>

File Manager

Save

Using web services of Campus A.



Note: All this work is shown only for Campus A. Similar way was adopted for all other campuses.

NAT configuration for Campus A:

Campus A

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#no ip nat inside
Router(config-if)#int se0/0/0
Router(config-if)#no ip nat outside
Router(config-if)#int se0/0/1
Router(config-if)#no ip nat outside
Router(config-if)#int se0/1/1
Router(config-if)#no ip nat outside
Router(config-if)#int se0/1/0
Router(config-if)#no ip nat outside
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat inside source static 200.100.192.10 192.168.1.3
Router(config)#ip nat inside source static 200.100.192.11 192.168.1.3
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

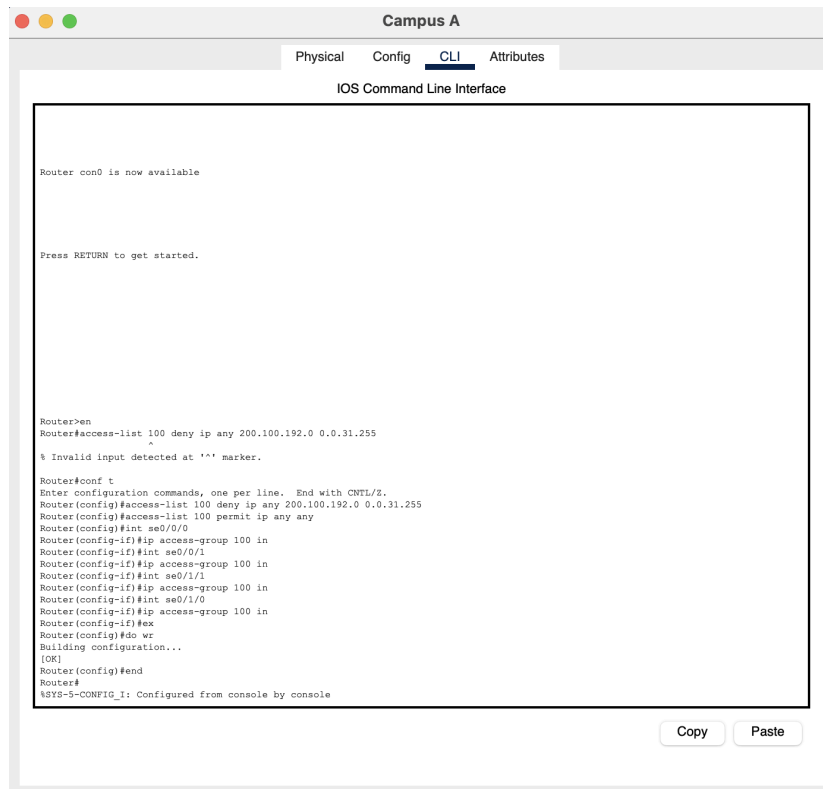
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0.10
Router(config-subif)#ip nat inside
Router(config-subif)#int se0/0/0
Router(config-if)#ip nat outside
Router(config-if)#int se0/0/1
Router(config-if)#ip nat outside
Router(config-if)#int se0/1/1
Router(config-if)#ip nat outside
Router(config-if)#int se0/1/0
Router(config-if)#ip nat outside
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

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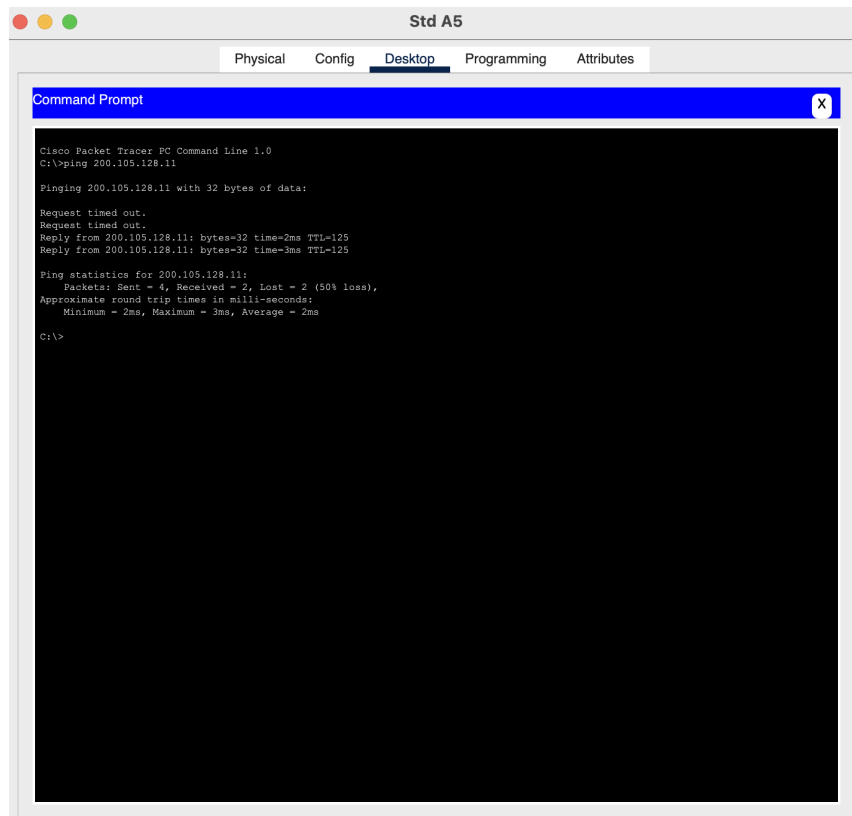
NAT table of Campus A:

NAT Table for Campus A				
Protocol	Inside Global	Inside Local	Outside Local	Outside Global
---	192.168.1.3	200.100.192.10	---	---
---	192.168.1.4	200.100.192.11	---	---

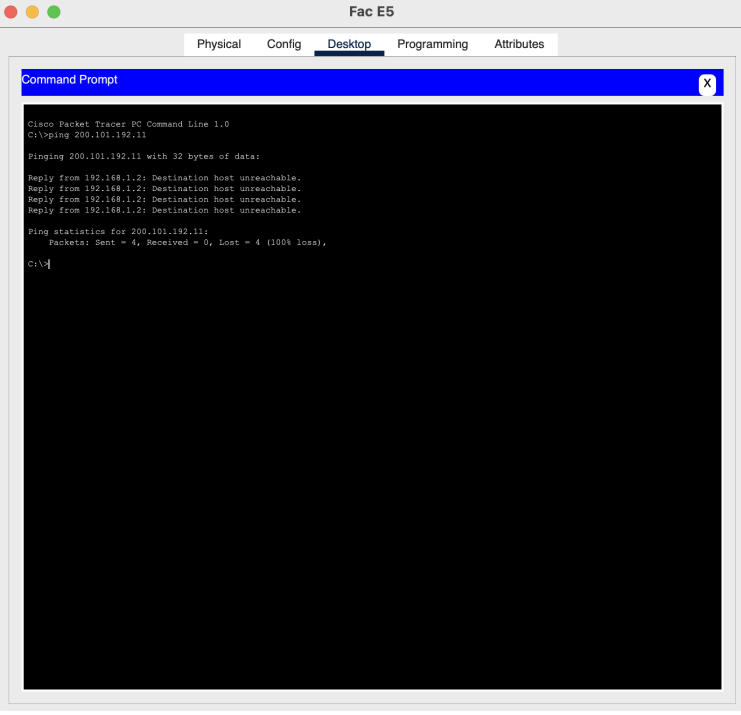
ACL for NAT of Campus A:



Sending packet from Students of Campus A to Faculty of Campus F:



Sending packet from Faculty of Campus E to Admin of Campus B using private address
(Failed):



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.101.192.11

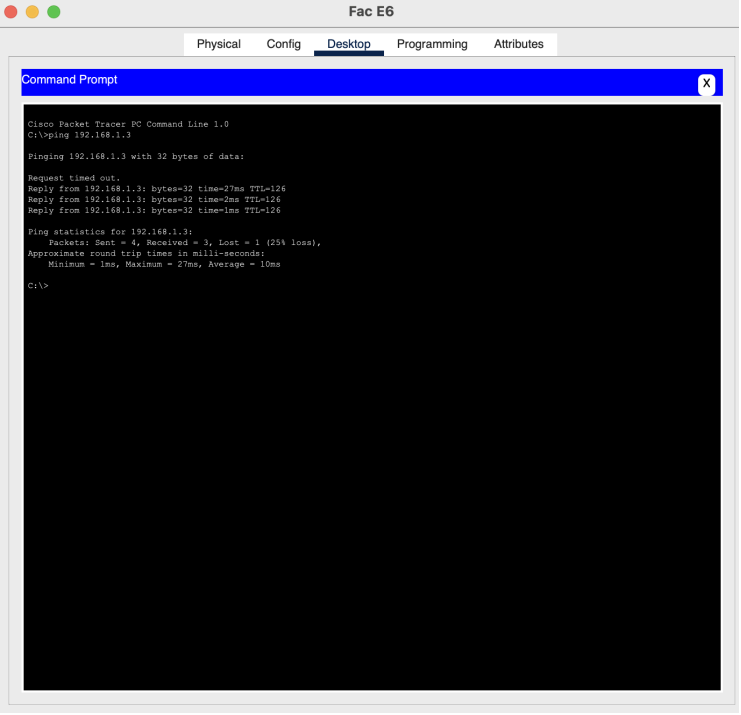
Pinging 200.101.192.11 with 32 bytes of data:

Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.
Reply from 192.168.1.2: Destination host unreachable.

Ping statistics for 200.101.192.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Sending packet from Faculty of Campus E to Admin of Campus B using public address
(Success):



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

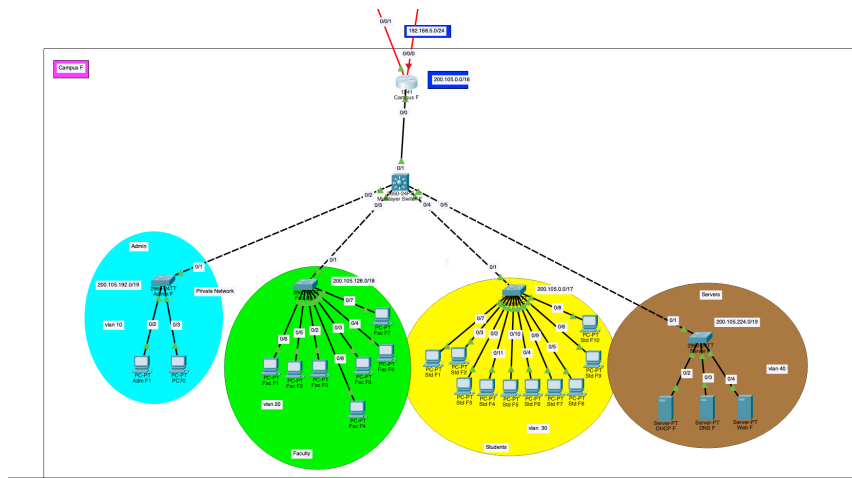
Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.3: bytes=32 time=27ms TTL=126
Reply from 192.168.1.3: bytes=32 time=2ms TTL=126
Reply from 192.168.1.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 27ms, Average = 10ms

C:\>
```

I shut down primary link and checked that secondary link working or not?



As the packets are sent, so secondary link is also working.

```
Fac F4
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.100.0.11

Pinging 200.100.0.11 with 32 bytes of data:

Request timed out.
Reply from 200.100.0.11: bytes=32 time=2ms TTL=125
Reply from 200.100.0.11: bytes=32 time=2ms TTL=125
Reply from 200.100.0.11: bytes=32 time=2ms TTL=125

Ping statistics for 200.100.0.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>
```